14/4/26

IA-2 - DDCC

Assignment - 2 :

(i) if a hypervisor introduces a 10% CPU, overhead, what is the effective CPU available to a VM running a workload that needs 4 cores on a 5-core physical CPU:

On a 5 core CPU, this leaves 90% of the total capacity for the VM :

$$5 \times (1 - 0.10) = 5 \times 0.9 = 4.5$$

The VM's workload requires 4 cores, which fits comfortably within the 4.5 cores effectively available after overhead.

(2) Overhead % due to virtualization 200%

$$= \frac{\text{VM latency} - \text{physical latency}}{\text{physical latency}} \times 100\%$$

$$= \frac{15\,ms - 5\,ms}{5\,ms} \times 100\%$$

$$= 10/5 \times 100\% = 200\%$$

The VM experiences 3x the latency of the bare metal sys resulting in a 200% overhead.

3) Maximum speedingup using 10 processors = 2.7027

(Exactly 100/37)

3. A program has 30% serial portion & 70% ||el

Calc max. speedup achievable using 10 processors:

$$S = \frac{1}{s + \frac{p}{n}} \Rightarrow \frac{1}{0.3 + \frac{0.7}{10}} = \frac{1}{0.3 + 0.07} = \frac{1}{0.37}$$

$$= 2.7027$$

Even with 10 processors, the serial portion limits the speedup to roughly 2.7 x.

4) A program with 25% serial portion runs on 40 pro. Calc scaled speedup. If the prm size doubled while keeping serial portion fixed what is the new scaled speedup?

This uses Gustafson's Law (Scaled speedup) which assumes scaled ||el portion of the work scales with the no. of processors while the serial portion remains fixed in abs time. Let $\alpha = 0.25$ be the serial fraction on a single processor for the Original pbm size.

$n = 40$

$$S(n) = \alpha + n(1 - \alpha)$$

$$= 0.25 + 40(0.75)$$

$$= 0.25 + 30$$

$$= 30.25$$

$$S(n) = n - \alpha(n-1)$$

$$= 40 - 0.25 \times 39$$

$$= 30.25$$

if pbm 2x, while serial portion fixed. The new total work is twice the Original:

$$\alpha' = \frac{0.25 \times \text{Original total}}{2 \times \text{Original total}} = 0.125$$

Applying Gustafson's law:

$$S'(40) = 0.125 + 40 \times (1 - 0.125)$$

$$= 0.125 + 35$$

$$= 35.125$$

This larger pbm size reduces the relative impact of the serial portion, improving the scaled speedup. closer to the ideal 40x. This illustrates why G.L is more optimistic than Amdhal's for real world workloads whr pbm size grow with available processors.

5) A "e" pgm runs on 16 processors & Achieves a speedup of 12. Using a Karp-Flatt metric Calc serial fraction (e). then if no. of processors ↑ to 32. but speedup improves Only to 18. recalc (e) & Analyse whats this indicates abt pgm Scalability:

The Karp-Flat metric estimates the effective serial func e from observed speedup(S) on p processors

$$e = \frac{\frac{1}{S(p)} - \frac{1}{p}}{1 - 1/p}$$

For $p=16, S=12$:

$$\frac{1}{12} - \frac{1}{16} = \frac{4-3}{48} = \frac{1}{48}$$

$$1 - \frac{1}{16} = \frac{15}{16}$$

$$e = \frac{1/48}{15/16} = \frac{1}{48} \times \frac{16}{15} = \frac{1}{45}$$

For $p=32, S=18$:

$$\frac{1}{18} - \frac{1}{32} = \frac{32-18}{18 \times 32}$$

$$\Rightarrow \frac{14}{576} = 7/288$$

$$1 - \frac{1}{32} = \frac{31}{32}$$

$$e = \frac{7/288}{31/32} = \frac{7}{279}$$

The slight ↑ in e (from 2.22% to 2.51%) as processors 2x indicates poor scalability. In a perfectly scalable pgm the effective serial fraction would remain constant (or ↓).

(6) In a system with 7 nodes, 2 nodes are faulty & can've arbitrarily (Byzantine faults). The nodes exchange msgs to reach agreement.

(i) what is the minimum no. of rounds of msg exchanges needed to guarantee consensus despite the faults, according to Lamport's Byzantine Generals Algorithm?

Lamport's Byzantine Generals (Oral-Msg) alg. generates consensus in a sync. system. despite up to f arbitary (Byzantine) faults if & only if the total no of nodes $n > 3f+1$

$n = 7$
$f = 2$ satisfies the bound exactly.

The alg. requires exactly $f+1$ r.of.msg exchange (the recursive Oral msg Alg OM(f) uses $f+1$ phases).

Thus, 2 faults require 3 rounds.

Each round take 100ms, total time $= 3 \times 100\,ms = 300\,ms$

7) Assume a distributed system has 10 nodes. with the prob. of each node becoming Byzantine faulty independently as 0.1 (10%)

(i) What is the prob. that atleast 3 nodes become Byzantine fault simultaneously?

(ii) If the system can tolerate upto 2 Byzantine faults. what is the faulty prob that it fails due to exceeding this threshold?

$$P(X=k) = \binom{10}{k} (0.1)^k (0.9)^{10-k}$$

$n = 10$; $p = 0.1$

$P(X \geq 3)$

Computed values: $P(X=0) \approx 0.3487$

$P(X=1) \approx 0.3874$

$P(X=2) \approx 0.1937$

$$P(X \geq 3) = 1 - [P(X)=0) + P(X=1) + P(X=2)]$$

$$= 0.0702$$

The system $\because$ fails with prob $= 7.02\%$ due to having 3 (or) more simultaneous Byzantine faults.